AlGaAs guided-wave second-harmonic generation at 2.23 μm from a quantum cascade laser

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We demonstrate the frequency doubling of a quantum cascade laser in a multilayered, partially oxidized GaAs/AlOx waveguide. Using the waveguide width to fulfill the phase-matching condition, the second harmonic is generated in the wavelength range between 2.2 and 2.4 μm, where not many semiconductor sources are commercially available to date. We discuss the impact of a few fabrication and experimental parameters on the conversion efficiency, an essential step toward the improvement and practical implementation of this proof-of-principle semiconductor microsystem.

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1. Introduction

The 2–4 μm wavelength range presents a large number of spectroscopic applications, for military as well as civil use [1]. The development of compact coherent sources operating at room temperature in this spectral range is, therefore, of great practical importance, several efforts being made to extend the emission ranges of laser diodes toward longer wavelengths [2] and quantum cascade lasers (QCLs) toward shorter wavelengths [3–5]. An alternative way to reach this frequency domain is guided-wave nonlinear frequency conversion. The AlGaAs platform is particularly suited for such nonlinear sources because of its mature technology, high nonresonant $\chi^{(2)}$, wide transparency range, and the possibility of electrical pumping. A few AlGaAs-based devices for frequency downconversion into the above spectral range have been demonstrated through different phase-matching (PM) schemes, spanning from an electrically pumped Bragg-mode parametric generator (modal PM) [6] to a near-IR (NIR) optical parametric oscillator in an AlGaAs/AlOx waveguide (form-birefringent PM) [7]. Although such guided-wave downconversion sources lend themselves to tunable operation, they suffer from high propagation losses.

For oxidized aluminum oxide (AlOx) waveguides, in particular, optical losses are highest at pump wavelengths of around 1 μm. However, at wavelengths longer than 1.1 μm, they are dominated by Rayleigh-like scattering [8] and become progressively smaller. It is thus interesting to reach the 2–4 μm range with an upconversion process in AlOx waveguides.

In addition, after several demonstrations at longer wavelengths [9–12], intracavity second-harmonic generation (SHG) has been reported in QCLs down to 2.25 μm [13–15], based on resonant $\chi^{(2)}$.

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To circumvent the design and fabrication constraints of intracavity SHG, a possible alternative is to address extra-cavity SHG in a semiconductor guided-wave device to be coupled to a QCL within a hybrid on-chip microsystem. This scheme is of course not restricted to one particular type of QCL, material, or wavelength range.

As an example of this strategy, we demonstrate quasi-continuous-wave (CW) SHG at around 2.23 μm of a QCL beam in GaAs/AlOx waveguides, whose fabrication technologies are well established. After a brief description of the frequency converter design and the fabrication process in Section 2, Section 3 is devoted to the linear and nonlinear waveguide characterization. Conclusions are drawn in Section 4.

2. Design and Fabrication

In the frequency doubling waveguide, PM relies on form birefringence, which is induced by alternating two materials with different refractive indices at a subwavelength scale [16]. Here, only two thin layers of AlOx suffice in the GaAs waveguide core to fulfill type-I PM, because the material dispersion between 4.46 and 2.23 μm is $\Delta n \sim 0.03$, i.e., considerably smaller than the typical value, $\Delta n \sim 0.14$, for down-conversion from 1.064 to 2.128 μm in similar devices with the same type-I PM condition [7]. Such reduction of the number of AlOx layers is interesting because their related defects are the main source of propagation losses in such waveguides [8]. Therefore, besides the perspective of a semiconductor microsystem combining the QCL and SHG sections in the same packaging, this work is motivated by the expectation of low losses in the passive nonlinear waveguide. A sketch of the nonlinear waveguide with the calculated field distributions of the interacting modes is displayed in Fig. 1. SHG takes place in the GaAs layers of the core (where $d_{14} \approx 90$ pm/V).

The following nominal structure has been grown on a GaAs substrate by molecular beam epitaxy (MBE): 5000 nm $Al_{0.8}Ga_{0.2}As$/430 nm GaAs/37.5 nm $Al_{0.98}Ga_{0.02}As$/741 nm GaAs/37.5 nm $Al_{0.98}Ga_{0.02}As$/430 nm GaAs/2000 nm $Al_{0.8}Ga_{0.2}As$/30 nm GaAs (cap layer). In order to ascertain the actual values of the composition and thicknesses of this thick multilayer, we measured the high-resolution x-ray diffraction spectrum (Brucker D8 Discover) around the (004) reflection (CuKα radiation).

By fitting it with a numerically simulated spectrum, we deduced the actual epitaxial structure, which differs by less than 3% from the nominal one: 4843.6 nm $Al_{0.81}Ga_{0.19}As$/416.8 nm GaAs/37.8 nm $Al_{0.94}Ga_{0.06}As$/718.3 nm GaAs/37.8 nm $Al_{0.94}Ga_{0.06}As$/416.8 nm GaAs/1937.5 nm $Al_{0.81}Ga_{0.19}As$/29.1 nm GaAs. The quality of this fit is apparent in Fig. 2, where the perfect agreement between the measured and simulated spectra ensures accuracy better than 10^{-2} on all layer parameters. This analysis is crucial to precisely calculate the PM condition because the latter is very sensitive to some growth parameters [17]. The epitaxial structure provides

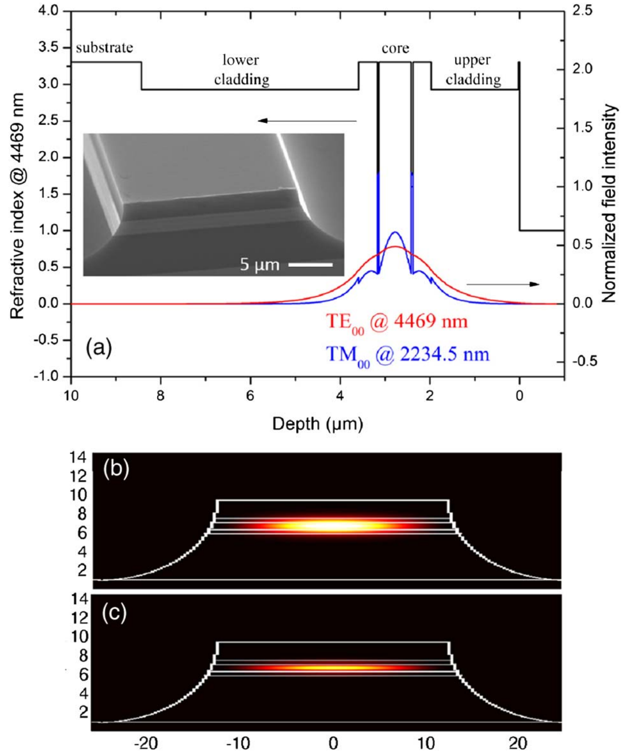

Fig. 1. Waveguide structure and mode intensity distributions at the FF and SH. (a) Vertical refractive-index profile (black) with the two low-index AlOx layers, as well as the TE_{00} mode at 4.469 μm (red) and the TM_{00} mode at 2.235 μm (blue); inset: scanning-electron-microscope picture of the sample. (b) 2D transverse profile of the TE_{00} mode at 4.469 μm. (c) 2D transverse profile of the TM_{00} mode at 2.235 μm.

the vertical confinement of the waveguide modes, whereas lateral confinement is provided by wet-etched 8 μm deep ridges oriented along the GaAs ⟨110⟩ direction. Since our monolithic QCL is not widely tunable, we need a coarse PM parameter other than wavelength to compensate for any fabrication tolerance. As 2D simulations showed that

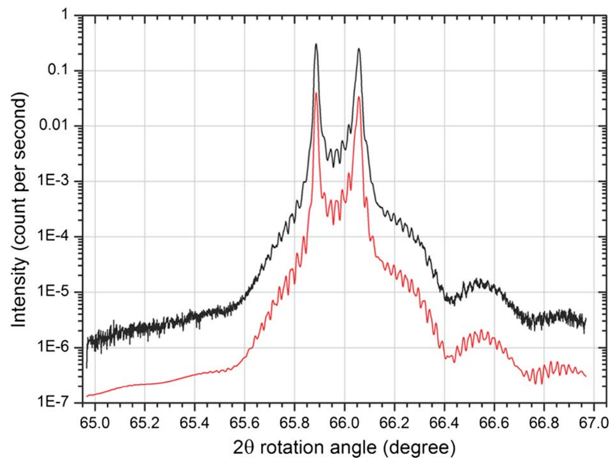

Fig. 2. High-resolution x-ray diffractogram of the MBE sample: experimental measurement (black curve) and simulated curve (red curve). The red curve has been translated vertically for sake of clarity.

waveguide width is a sensitive parameter, we chose to use it to achieve PM. The resulting tunability for the as-grown sample is displayed in Fig. 3. We, therefore, fabricated several samples with waveguide widths ranging from 10 to 35 μm.

The linear patterns were drawn on the resist deposited on our sample by e-beam lithography, preferred to its optical counterpart for its greater design versatility. Then we deeply wet-etched the waveguides with a CH_3COOH:HBr:$K_2Cr_2O_7$ (1:1:1) solution. After cleaving 1–3 mm long samples and carefully calibrating the oxidation furnace, the $Al_{0.98}Ga_{0.02}As$ layers were selectively oxidized in AlOx over ~35 min at 430°C in a wet atmosphere obtained by passing a N_2 gas flow through a water bubbler at 70°C [18]. The typical aspect of a waveguide is displayed in the inset in Fig. 1(a).

3. Experiment and Discussion

For the measurements, we used two laser sources. The first one is a distributed-feedback (DFB) QCL fabricated by the Alcatel Thales III-V Lab. It emits the fundamental frequency (FF) pump beam at a wavelength of around 4.47 μm in a pulsed regime. Such a laser is based on InGaAs/AlInAs/InP strain-compensated growth. The ridge is chemically etched by the double-trench technology and is electroplated for efficient thermal management. For the same reason, it is mounted epi-down on an AlN carrier. The second one, used for alignment purposes, is a commercial Nanoplus DFB laser diode emitting CW at around 2.12 μm. We coupled either of the laser beams into the waveguide with a ZnSe objective. A similar objective was used for collection and collimation of the output at both the FF and the second harmonic (SH). After the output objective, two alternative analysis paths were used, depending on the wavelength to be collected. The FF throughput was directed to an ultrafast detector (VIGO System, model PCI-21E-12/VPAC-1000F). The NIR radiation was focused with a BK7 lens on an InGaAs photodiode (Judson Technologies) with a cut-off at 2.4 μm. The photocurrent was then measured with a low-noise transimpedance amplifier followed by a lock-in (Princeton Applied Research, model 5210). To monitor the injection of the beams into the waveguide, we also had a quantum-well infrared photodetector (QWIP) camera, whose detection range contains all the wavelengths used in the experiment. The setup is sketched in Fig. 4.

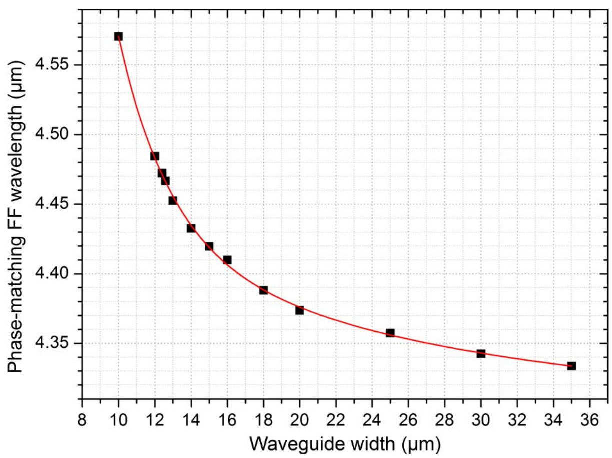

Fig. 3. PM wavelength versus waveguide width. The dots are calculated for n(AlOx) = 1.6, whereas the red line is a mere guide to the eyes. Given the fabrication tolerance on n(AlOx), this curve only provides a first useful indication of the PM condition.

In the nonlinear experiment, we used 0.3 μs QCL pulses to keep the wavelength chirp of the pulse as low as possible. In order to measure guided-wave losses, α_ω, conversely, we exploited the larger chirp associated to 3 μs long pulses to apply the Fabry–Perot method [19] at around 4.5 μm. Although the α_ω measured were often between 2 and 3 cm^{-1}, we could find some waveguides with $\alpha_\omega < 1$ cm^{-1}. Such data dispersion stems from the multimode nature of our waveguides, whose losses can be overestimated if the laser beam is not injected properly into the transverse fundamental mode [19]. We can, therefore, estimate α_ω to be around 1 cm^{-1}. These mid-IR losses are lower than those measured by Ravaro *et al.* in a similar waveguide [20], thanks to the smaller number of AlOx layers in our sample. The low level of losses also confirms that the laser wavelength is beyond any tail of the AlOx absorption peak around 3.3 μm as observed in [20].

After careful alignment of a 1.5 mm long sample with 14–21 μm wide waveguides, and of the SH detection path, we only detected SH power out of waveguides with widths between 14 and 15.5 μm. In addition to the blindness of the InGaAs photodiode and the BK7 lens to the QCL radiation, we used a Glan–Taylor polarizer after the collection objective to verify the SHG polarization rule: we oriented it to select only the TM-polarized radiation. To modulate the effective injected power without changing the QCL injection parameters, we used another grid polarizer suited for the QCL wavelength.

The dependence of SH power on the TE_{00} mode-injected pump power at the entrance of the waveguide is displayed in Fig. 5. These measurements

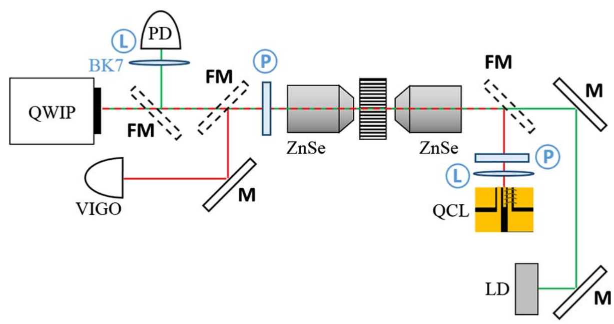

Fig. 4. Setup for waveguide characterization. The key element of this setup is the QWIP camera, which allows monitoring the injection of the QCL beam into the waveguide. The green (red) line highlights the path of the near- (mid-) IR beam. M, standard mirror; FM, flip mirror; L, lens; P, polarizer; LD, laser diode; PD, photodiode.

were performed in a 15.3 μm wide waveguide where we obtained the highest output power. We obtained a maximum external SH power of 22 nW, for a 53.65 mW peak value of the FF external power. The internal SH power was estimated through calibration of the optics and accurate 2D-FDFD (finite difference frequency domain) calculation of the modal reflectivity at the waveguide end facet. Correspondingly, the effective FF internal power was estimated using Malus' law for polarization (as the QCL is polarized linearly, the FF power injected into the TE_{00} guided mode scales as $\cos^4\theta$, where θ is the polarizer angle), and by calculating the overlap between the focused Gaussian beam and the TE_{00} mode after characterization of the injection objective. Note that the internal pump power is the peak value, whereas the 300 ns pulse of the QCL acts as a 3% duty-cycle quasi-CW radiation in the 1.5 mm long waveguide. The theoretical relation of the generated SH power with the FF pump power is given by $P_{\mathrm{SHG}} = \eta P_p^2$, with [21]

$$\eta = \eta^0 L^2 e^{-(\alpha_\omega + \alpha_{2\omega}/2)L} \times \frac{\sin^2\left(\frac{\Delta k L}{2}\right) + \sinh^2\Gamma}{\left(\frac{\Delta k L}{2}\right)^2 + \Gamma^2}, \quad (1)$$

where $\alpha_\omega(\alpha_{2\omega})$ stands for the guided mode losses at frequency $\omega(2\omega)$, Γ for $(\alpha_\omega - \alpha_{2\omega}/2)(L/2)$, L for waveguide length, Δk for phase mismatch, and η^0 for the normalized conversion efficiency.

The good agreement between the fits and the experimental points confirms that we observed SHG in our waveguides. However, we measured the frequency-doubled signal in wider guides than those predicted in Fig. 3. This might be due to the sensitivity of the phase-matched width to the AlOx refractive index [17]. The AlOx refractive index is indeed strongly dependent on the oxidation conditions, and is not entirely well known in our structure. By adjusting this parameter within the range mentioned

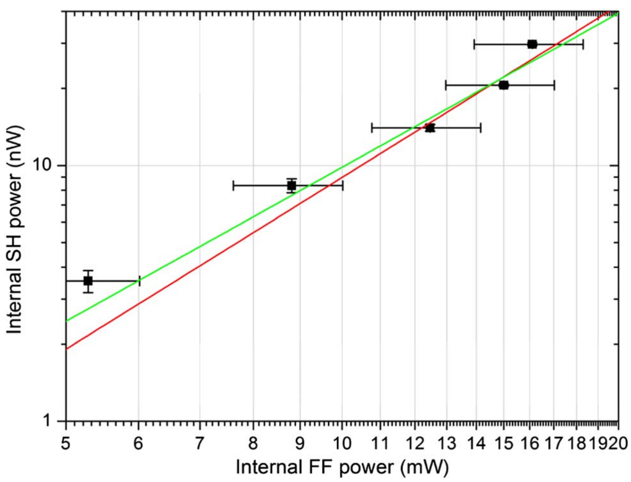

Fig. 5. Internal pump power versus the FF power coupled into the TE_{00} mode. The dots are experimental points. The red line is the best $y = ax^b$ fit whose output is $b = 2.2$ and the green line is the same fit with $b = 2$, from which we extract the efficiency of our process.

by De Rossi *et al.* [17], the predicted PM waveguide width varies from 9.5 to 25 μm.

Let us now use Eq. (1) to both infer the experimental SHG efficiency, η/L^2, of our waveguides and to calculate its theoretical counterpart. From the quadratic fit of Fig. 5, we can extract the former as 0.44% $W^{-1}\,cm^{-2}$. For perfect PM and assuming guided mode losses of around 1 cm^{-1} at both the FF and SH, the calculated value of such efficiency is 92% $W^{-1}\,cm^{-2}$. As the multimodality of the waveguide prevented us from measuring reliable values of the losses at the SH frequency, we estimated them from our experience with similar waveguides [8].

Such two-order-of-magnitude discrepancy mainly stems from the contingent fact that the growths of the QCL and the AlOx waveguide were not matched perfectly and, as a consequence, we could not achieve PM in single-mode waveguides. In practice, the resolution of the QWIP camera being insufficient to distinguish the order of the injected mode, we could not avoid exciting higher-order, phase-mismatched FF modes besides the fundamental one. Although a fraction of the injected light clearly excites the TE_{00} mode (because PM is only possible between the two fundamental modes), we can only calculate the upper value of this fraction. This results in an overestimation of the FF power in the fundamental mode, which in turn leads to underestimation of the SHG efficiency.

In addition, we could not search for the PM condition continuously, because the only accessible PM adjustment parameter—the ridge width—is coarse and discrete; thus, we cannot prove that PM was met perfectly. However, the small dispersion of GaAs in this wavelength range and the small discretization step for the waveguide width suggest that the latter should not constitute the limiting factor.

4. Conclusion and Perspectives

In this paper, we have reported a proof-of-principle SHG from a QCL beam in a GaAs/AlOx waveguide at room temperature. The separation between the laser and the frequency converter allows searching the PM condition independently of the laser wavelength.

We presently plan to grow an optimum heterostructure in order to fulfill the PM condition in single-mode waveguides. By doing this, the selective injection into the waveguide will be greatly ameliorated, with a strong expected impact on the experimental conversion efficiency. In addition to this practical improvement, an optimum design based on the specific QCL used here (including a third AlOx layer and a ridge width of 5 μm) will also enhance the nonlinear overlap integral from 1.54×10^{-5} to 2.65×10^{-5} V^{-1}, with an additional factor-3 increase of the calculated conversion efficiency.

The small sizes of the QCL and the GaAs/AlOx waveguide lend themselves to the compact packaging of an electrically injected coherent source at 2.23 μm, and one can even envisage hybridizing them on the same semiconductor chip [22]. Finally, for a fully monolithic integration, the InP platform seems an interesting longer-term scenario for a nonlinear active photonic chip.

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